



Cartography M.Sc.

From Data to Interventions: A Visual Analytics Framework for Agroforestry Management

Case study: Living Labs in the Greater Northern African Region

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1 Motivation and Problem Statement

Land degradation represents one of the most significant global challenges in the present time, affecting ecological stability, food security, and climate resilience. Beyond its environmental consequences, it also has economic implications, with annual losses estimated at 10–17% of global Gross Domestic Product. Land degradation reduces water retention capacity and increases vulnerability to droughts and floods, affecting more than 3.2 billion people worldwide (Montanarella et al., 2018). At the same time, only 24% of the land around the world is considered suitable for cultivation (AbdelRahman, 2023), leaving little margin for further losses.

A key driver of land degradation is the mismatch between land use practices and local environmental conditions, where land is used in ways that are neither efficient nor appropriate for its natural capacity. One of the most widely recognized approaches to addressing this challenge is agroforestry, which has strong potential to restore soil health, improve climate resilience, and maintain biodiversity (Dinesha et al., 2023; Bhandari et al., 2023). Agroforestry is particularly relevant in arid and semi-arid regions, where land degradation is intensifying over time. This urgency is increasingly acknowledged at the international level through research programmes and policy initiatives that promote agroforestry as a strategy for sustainable land management. Among these, the EU Horizon Europe TRANS-SAHARA project applies the Water-Energy-Food-Ecosystems (WEFE) Nexus approach to co-design and validate agroforestry interventions in Living Labs across the Greater Northern African Region (TRANS-SAHARA Consortium, 2025).

However, translating this international commitment into practice requires tools that make environmental data interpretable and actionable for the practitioners who implement these interventions. The initial assessment of land suitability is particularly challenging, as it requires the integration of multiple environmental factors through

geospatial analysis while producing outputs that are clear and usable in practice. Existing digital tools are often designed for specialists and remain insufficiently intuitive or accessible for practitioners (Terribile et al., 2024). Those tools are characterized by limited functionality, complex interfaces, and deliver static outputs or maps without interactive or spatial decision support features (Tranchina et al., 2024). This creates a gap between the availability of environmental data and its practical applicability.

Cartographic design offers one way to bridge this gap. Cartographic methods assist in the translation of complex environmental data into structured visual representations that are intuitive representations that are more intuitive and accessible (MacEachren, 1995). Furthermore, modern web cartography allows users to interact dynamically with spatial information and can therefore support and simplify decision-making processes (Kraak and Ormeling, 2020). In this context, cartographic design acts as an instrument for linking analytical complexity with practical land management decisions.

2 Research Identification

This thesis is embedded in TRANS-SAHARA, an EU-funded project that develops WEFE-Nexus-based approaches to agroforestry management in the Greater Northern African Region. The project uses water security as a central entry point for designing and managing agroforestry systems, and it validates technical and non-technical solutions in Living Labs in Tunisia, Ghana, and Ethiopia. Within this broader framework, the present thesis contributes to the development of the Intervention Design Tool by focusing specifically on its cartographic interface for spatial planning and decision support.

Agroforestry and WEFE-Nexus intervention planning in Living Labs requires decisions that are inherently spatial, multi-criteria, and stakeholder-dependent. Interventions such as agroforestry system placement, managed aquifer recharge structures, wells, solar-powered irrigation facilities, water-harvesting measures, and ecosystem protection zones must be located, configured, and compared in relation to environmental constraints, land suitability, local priorities, and future scenarios. In the TRANS-SAHARA context, these decisions are expected to be supported by the Intervention Design Tool, which integrates modelling and data analysis to inform intervention planning.

This work aims to develop a **prototype of a map-driven, web-based Intervention Design Tool module for agroforestry management** for the Living Labs of the Great North African Region, exploring cartographic methods as the primary instrument for communicating multicriteria land suitability analysis.

The tool will enable practitioners to define an Area of Interest (AOI) and explore interactive spatial outputs representing land suitability for different agroforestry interventions. Research examines how cartographic design principles can transform complex environmental datasets into clear actionable information for practitioners.

Research Objective 1

To design a cartographic, map-based interface for planning agroforestry and WEFE-Nexus interventions in Living Lab environments, based on stakeholder requirements

and the spatial characteristics of interventions, constraints, suitability factors, and scenarios.

Research Objective 2

To empirically evaluate how different cartographic representation and interaction strategies support stakeholders in planning, comparing, and revising interventions, and to derive design principles for map-based intervention planning interfaces.

2.1 Research Questions

Research Question 1.1 What cartographic and interaction design requirements do domain experts identify for a map-based agroforestry intervention planning interface in Living Lab environments?

Research Question 1.2 How can cartographic visual variables, symbology strategies, and interaction operators be designed to represent intervention categories, suitability factors, spatial constraints, and scenarios in an intuitive planning interface?

Research Question 2.1 To what extent do the cartographic representation and interaction strategies of the developed prototype support stakeholders in planning agroforestry interventions?

Research Question 2.2 What cartographic design principles for map-based intervention planning can be derived from empirical evaluation with domain experts?

2.2 Innovation Intended

The intended innovation of this thesis is not the creation of another generic web map or dashboard, but the development of a cartography-based planning interface concept tailored to spatial intervention design in agroforestry and WEF-Nexus contexts. Its contribution lies in translating complex intervention planning requirements into a coherent cartographic workflow that supports spatial decision-making across heterogeneous intervention types and user groups.

Scientifically, the innovation is expected in three areas. First, the thesis will propose a structured way to represent different intervention geometries and planning entities within one interface, including point-, line-, and area-based interventions. Second, it will address the cartographic challenge of integrating intervention options with constraints, suitability information, and scenario comparisons in a planning-oriented map environment. Third, through empirical validation, it will generate transferable design knowledge on how map-based interfaces can better support intervention planning in Living Labs and similar spatial decision-support contexts.

2.3 Related Work

2.3.1 Agroforestry Digital Tools and Decision Support Systems

Existing digital tools for agriculture, such as AgriSens, primarily oriented towards smart farming support and crop productivity optimisation. One of the most advanced examples

in this field is OverYield, which allows users to plan plantings and supports agronomic advising and economic forecasting for perennial agricultural systems.

A further category of platforms focuses on analysing environmental data to produce economic reports on land suitability and field characteristics. For example, CourageousLand offers automated land diagnostics based on user-defined field boundaries, covering planning, financing, and carbon sales. However, the platform does not provide cartographic representations, and its diagnostic functionality is geographically limited to Brazil. Common to both categories is the limited availability of interactive cartographic outputs for practitioners, as well as geographic restrictions.

Compared to commercial tools, academic studies focus more on methodology and environmental analysis, thereby providing a stronger scientific basis for agroforestry planning. One of the earliest attempts to link environmental datasets with user-defined inputs for agroforestry planning was made by Ellis et al., 2000. The FADSS (Florida Agroforestry Decision Support System) combines soil and climate data with tree species characteristics and user-defined locations to suggest best planting decisions.

Recent studies have focused more on algorithmic improvements and the integration of environmental and socioeconomic data. Yadav et al., 2024 developed a prototype of decision support system for tree selection and plantation system for tree selection and plantation planning in subtropical India, integrating ecological, social, and agro-economic criteria. This work highlights the importance of incorporating stakeholders into the planning process. Similarly, Gosme et al., 2025 introduced AgroforesTreeAdvice in collaboration with the European Agroforestry Federation, integrating eight existing European tree selection tools into a unified interface through a hierarchical criteria framework. AgroforesTreeAdvice allows users to input local parameters such as soil type, climate, and production goals to receive species recommendations. Despite its strong knowledge base, the tool does not offer spatial cartographic interpretation.

2.3.2 Spatial Suitability Modelling and Environmental Analysis

The methodological basis for the assessment of land suitability for agroforestry is based on two approaches: multicriteria spatial analysis and machine learning. A number of studies have demonstrated the effectiveness of GIS-based multicriteria analysis for land evaluation (Wotlolan et al., 2021; Boggia et al., 2018; Ozkiper et al., 2024). For instance, Wotlolan et al., 2021 applied an integrated GIS-MCDA framework combining heterogeneous environmental datasets to evaluate land suitability for multiple crop categories in Sigatoka Valley, Fiji. Their study highlights the analytical value of integrating multiple spatial factors within a single suitability model.

Machine learning has been proposed as a further methodological advance. Singh et al., 2022 extended the multicriteria approach by incorporating ML algorithms, demonstrating improved applicability in heterogeneous landscapes. However, this depends on the availability of existing local agroforestry practice data and is therefore constrained by data availability and geographical specificity.

At a broader institutional scale, Terribile et al., 2024 developed LANDSUPPORT web-based platform, which combines environmental datasets with an analytical engine to evaluate trade-offs between land uses for policy-makers. Platform demonstrates the

potential of integrated environmental analysis, but outputs remain technically complex and difficult to interpret without specialist knowledge.

2.3.3 User-Centred design and usability in geo-information tools

Cartographic visualization is known to make complex natural and environmental data more accessible and spatially relevant (Walker et al., 2020; Cao and Chen, 2024). According to Movik et al., 2021, cartographic visualization not only represents reality, but also plays an active role in shaping interpretation and supporting decision-making

Following this statement, Lee et al., 2022 developed a decision support system and visual language for representing complex natural processes, including hydrological water flow systems and environmental simulations, in ways that are more understandable to practitioners. Their work demonstrated improved communication between the public and domain experts in planning processes.

Similarly, Walker et al., 2020 introduced an Interactive Catchment Explorer for visualising spatial patterns in environmental datasets through cross-filters, multivariate criteria, and coordinated map views. Research showed that interactive visualization fosters stakeholder collaboration and creates an iterative feedback loop between environmental monitoring, modeling and decision-making.

Recent studies represents that existing agroforestry tools often focus on species selection, productivity optimisation, or environmental diagnostics, while offering limited support for interactive spatial planning. In contrast, geoinformation tools provide advanced visualisation and analytical capabilities, but are rarely adopted to the specific requirements of agroforestry intervention planning. As a result, there remains a need for user-friendly cartographic interfaces that translate suitability analysis and spatial constraints into actionable planning support.

3 Project Setup

3.1 Scope

This thesis focuses on the design and validation of a map-based cartographic interface that contributes to the TRANS-SAHARA Intervention Design Tool. The work will support the planning of spatially explicit agroforestry and WEFE-related interventions in Living Lab settings. Relevant intervention types may include agroforestry system layouts, tree-based restoration areas, contour strips, riparian buffers, managed aquifer recharge measures, wells, farm ponds, water-harvesting structures, and solar-powered irrigation or pumping facilities.

The scope is intentionally limited to the cartographic and interaction layer of the tool. The thesis will not develop the WEFE, agroforestry, or optimization models themselves, nor will it depend on the complete availability of final project data. Instead, it will develop a data-agnostic but domain-grounded interface concept based on stakeholder requirements, realistic planning tasks, and representative intervention scenarios.

3.2 Methods Adopted

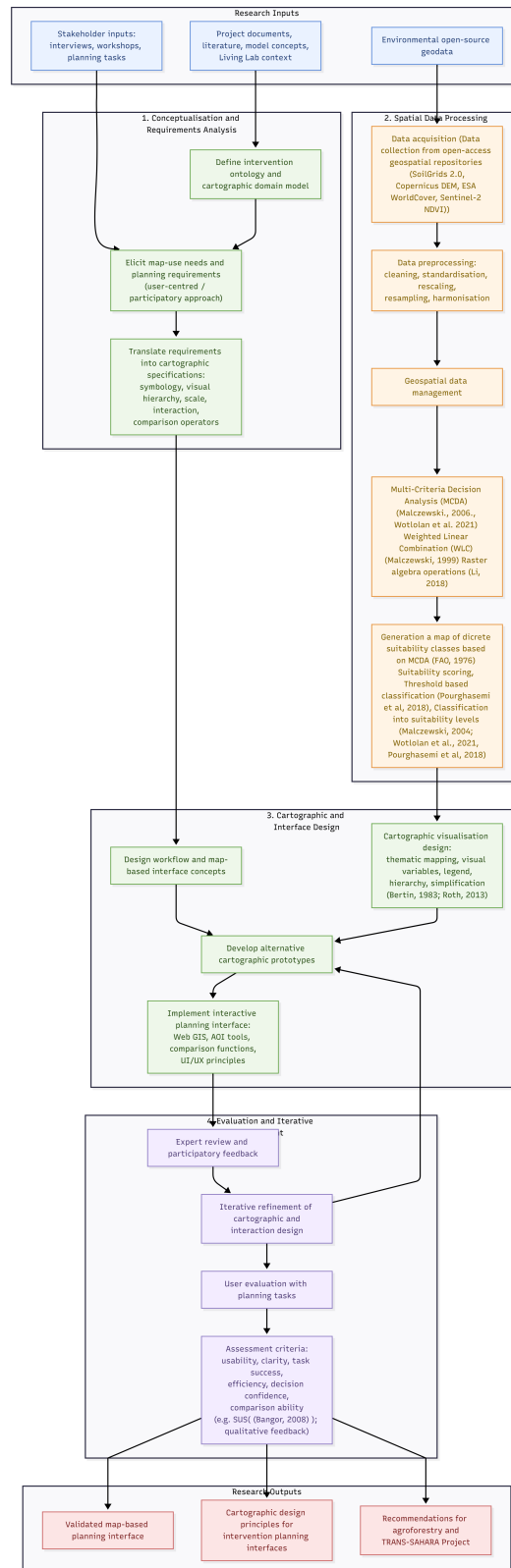


Figure 1: Cartographic workflow for the development and evaluation of the TRANS-SAHARA intervention planning interface

The methodological workflow of this thesis is structured into several interconnected blocks, illustrated in Figure 1. In the workflow, blue blocks indicate research inputs, green blocks represent design and development activities, orange blocks denote spatial analytical methods, purple blocks correspond to evaluation stages, and red blocks indicate research outputs.

The first block, *Conceptualisation and Requirements Analysis*, is based on the review of project documents, scientific literature, and the Living Lab context. These sources will be used to define the intervention ontology and the cartographic domain model, as well as to identify user needs and planning requirements relevant to agroforestry and WEFE-Nexus intervention planning.

The second block, *Spatial Data Processing and Suitability Analysis*, follows a standard geospatial workflow. Environmental datasets from open-access repositories will be collected, harmonised, and organised in a geodatabase. The suitability map will be generated using Multi-Criteria Decision Analysis (MCDA). Raster-based environmental criteria will be standardised, weighted, and combined to calculate suitability scores (Malczewski, 2004; Li et al., 2017). In addition, threshold-based classification will be applied to convert continuous scores into discrete suitability classes (Pourghasemi and Rossi, 2018).

The third block, *Cartographic and Interface Design*, applicable cartographic design principles, including thematic mapping, visual variables, legend design, visual hierarchy, and map simplification will be followed design principles established by Bertin, 1983 and Roth, 2013.

The final block consists of the *Evaluation and Iterative Refinement* of the developed prototype. The interface will be reviewed by experts involved in the TRANS-SAHARA project, and the collected feedback will be used to improve the cartographic representation and interaction design. The evaluation will combine qualitative feedback with standardised usability instruments such as the System Usability Scale (SUS) (Bangor et al., 2008).

3.3 Planned Schedule of the Project

The planned schedule of the project can be seen on this page. In the current plan, time is distributed between the following blocks: literature review, suitability analysis, cartographic interface design, stakeholder evaluation, and thesis writing. Meetings with supervisor will be scheduled each few weeks in person in EURAC Research Center. Practical part of the thesis will be completed by the mid-July. Evaluation workshop scheduled from last weeks of July to mid-August. In parallel with the practical phases of the project, thesis chapters will be drafted and submitted for review progressively.

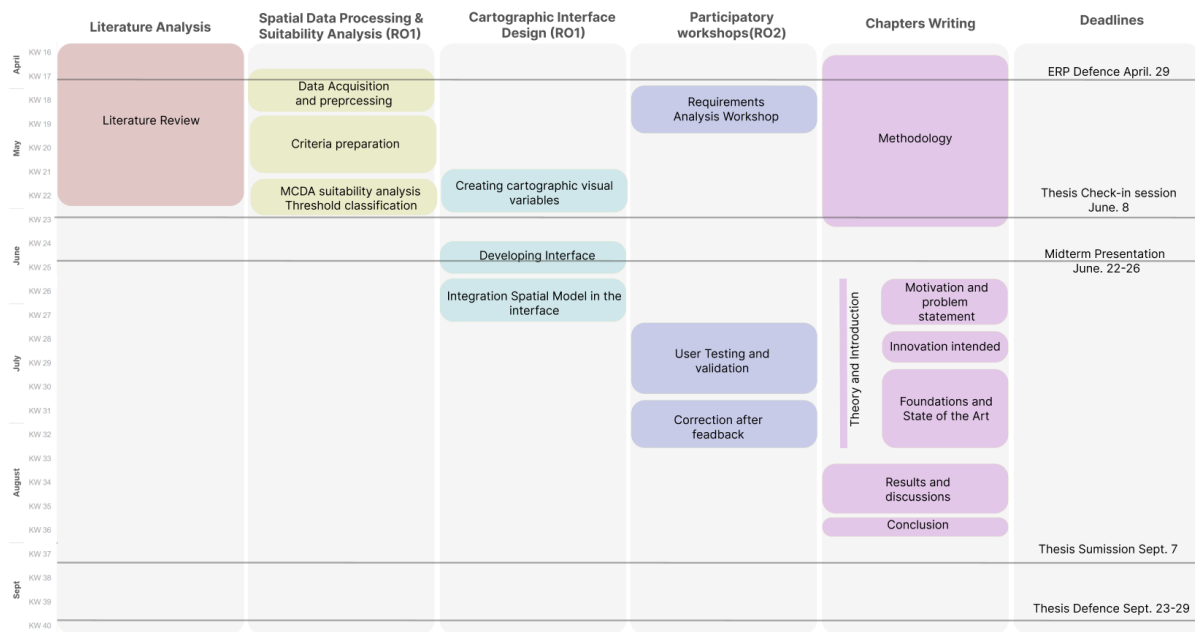


Figure 2: Planned project schedule

3.4 Risks and Contingencies

Risk	Probability	Contingency
Technical difficulties in geopro-cessing workflows	Low	Rely on established GIS libraries and tested workflows (QGIS, Python); prioritise core functions
Low participation of TRANS-SAHARA experts	Low	Organise workshops in flexible for-mats and time (online/in-person), establish early communication with participants
Limited functionality or integration issues in the web-based module	Medium	Prioritise core functionalities (AOI selection, suitability output, visual-isation); use well-supported open-source libraries
Low usability or poor data interop-erability of the interface	Medium	Apply established cartographic de-sign principles; validate interface decisions through Workshop
High predictive uncertainty of en-vironmental data in arid and semi-arid regions	High	Assess and document uncertainty layers prior to multicriteria analysis; use proxy variables
Missing or incomplete environmen-tal data in the study area	High	Assess data availability before anal-ysis; apply proxy variables where applicable and document all substi-tutions

Table 1: Risk assessment ordered by probability level, from lowest to highest

4 Resources Required

The project requires access to open-source datasets and their technical documentation, relevant information from the Living Labs, and domain experts for the validation phase.

- **Data:** Relevant datasets include soil data from SoilGrids 2.0 (e.g. soil texture, organic carbon, pH, and depth), land cover data from the Copernicus Land Monitoring Service and satellite imagery for manual classification, climate data from the Copernicus Climate Data Store and WorldClim, hydrological data from HydroSHEDS or similar sources, and topographic data from Copernicus DEM (elevation, slope, and aspect).
- **People:** The project requires access to experts from the TRANS-SAHARA consortium for feedback and validation; experts from EURAC Research, the Center for Sensing Solutions; the TUM Chair of Urban Water Systems Engineering, and the TUM Chair of Land Management.
- **Software and hardware:** GIS Software (GIS, QGIS), programming tools (Python with geospatial libraries), JavaScript frameworks, web mapping frameworks (Leaflet or Mapbox). No specialised hardware beyond a standard computer workstation is anticipated.
- **Finances:** No significant financial resources are required.

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Acronyms

WLC	Weighted Linear Combination
MCDA	Multi-Criteria Decision Analysis
SUS	System Usability Scale
GIS	Geographic Information System
AOI	Area of Interest

Declaration of Language Assistance

In the preparation of this proposal, language assistance tools, including OpenAI ChatGPT and Claude, were used solely for the purposes of grammar correction and stylistic improvement.